

California State University, East Bay

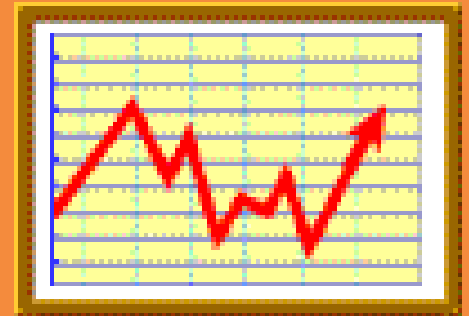
College of Business and Economics

BAN 673 Time Series Analytics

Time Series Data and Data Patterns



Lecture Materials



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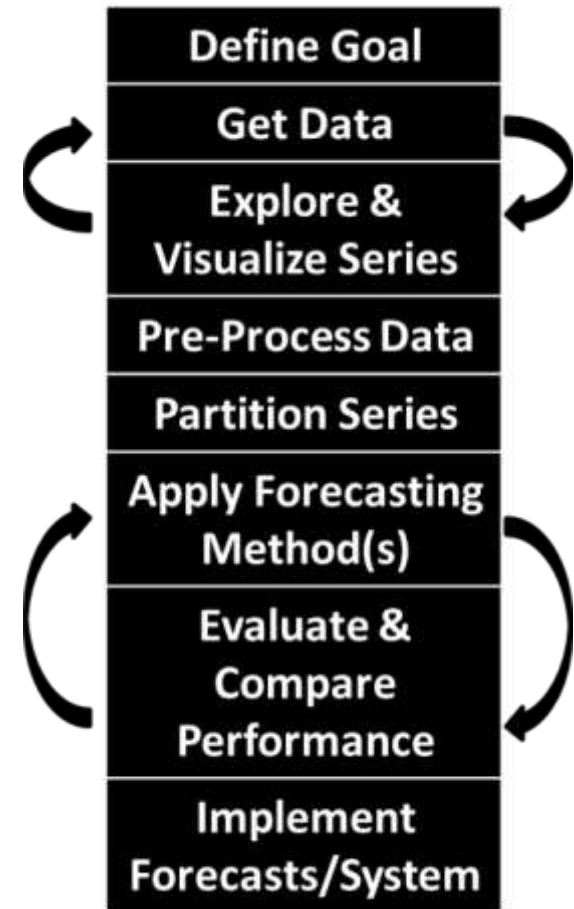
Lecture Objectives

- Define data collection step in time series forecasting
- Explain data quality, temporal frequency, series granularity, and domain expertise in time series data
- Explore time series data by identifying time series components: trend, seasonality, level and noise
- Use R to identify time series components
- Apply autocorrelation to identify time series components
- Build *correlogram* (autocorrelation function – *Acf*) and recognize time series components
- Identify time series components using *Acf* in R
- Visualize time series using charts in R
- Utilize pre-processing step in time series forecasting



Step 2: Data Collection (Get Data)

- **Data collection** should consider various aspects of forecasting goal (step 1)
- Considerations at the data collection level which can affect the forecasting results:
 - Data quality
 - Temporal frequency
 - Series granularity
 - Domain expertise



Considerations in Data Collection

■ *Data Quality*

- Specifically important in time series forecasting, where *sample size is relatively small*, typically no more than a few hundred values in a series
- With multiple sources, data collection may be *done from different departments*, so compare and choose the best data to apply
- For forecasting, *additional data may be collected in the future* from the same source
- External data may be also required to do a more accurate forecasting

■ *Temporal Frequency*

- Consider *frequency of time series the same as the frequency of the required forecast* (from the forecast goal)
 - » If the goal to forecast is next day energy usage in a building, then minute-by-minute or hourly day is less useful than daily aggregates of historical data
 - » High frequency data may have more noise, while aggregated data will reduce it
- Sometime: *use aggregated data, model aggregated forecasts, and then disaggregate* the forecasting results

Considerations in Data Collection (continued)

■ **Series Granularity**

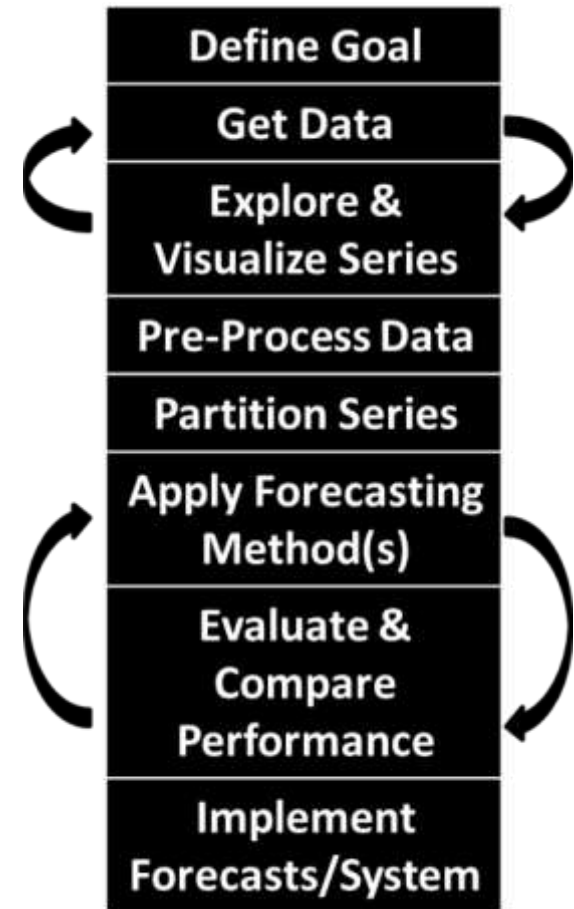
- *Granularity refers to the coverage of the data*
 - » In terms of geographical area, population, time of operations, etc.
 - » Depending on the forecast goal, series granularity may require different depth, e.g., forecasting general population growth vs. forecast of population of various age groups, states, region, etc.
- Level of granularity needs to be *aligned with the forecasting goal*
- Exploring *different aggregation and data slicing* levels are often needed to obtain appropriate time series

■ **Domain Expertise**

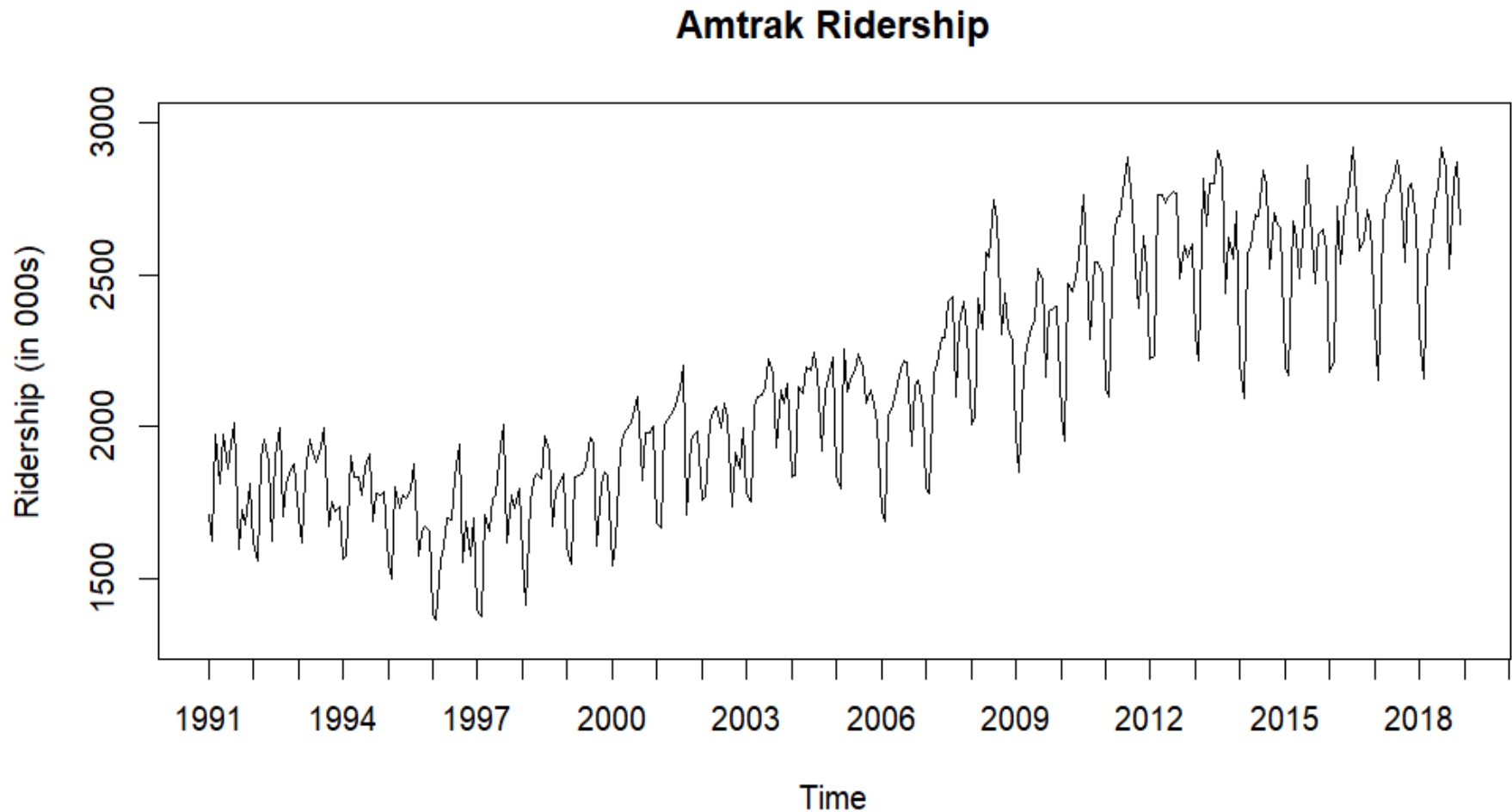
- Domain expertise (“art of forecasting”) is *needed to identify which data to employ* (daily vs. hourly, how far back, from which source, etc.)
- Help to evaluate practical implications of forecasting results
- Requires communication and coordination between business data analysts that perform forecasting and domain experts

Step 3: Explore & Visualize Series

- ***Explore time series*** step is used to do preliminary time series analysis of the data, and in particular
 - Identify *data patterns (components)* of the historical time series data
 - Use *autocorrelation* to investigate various data patterns



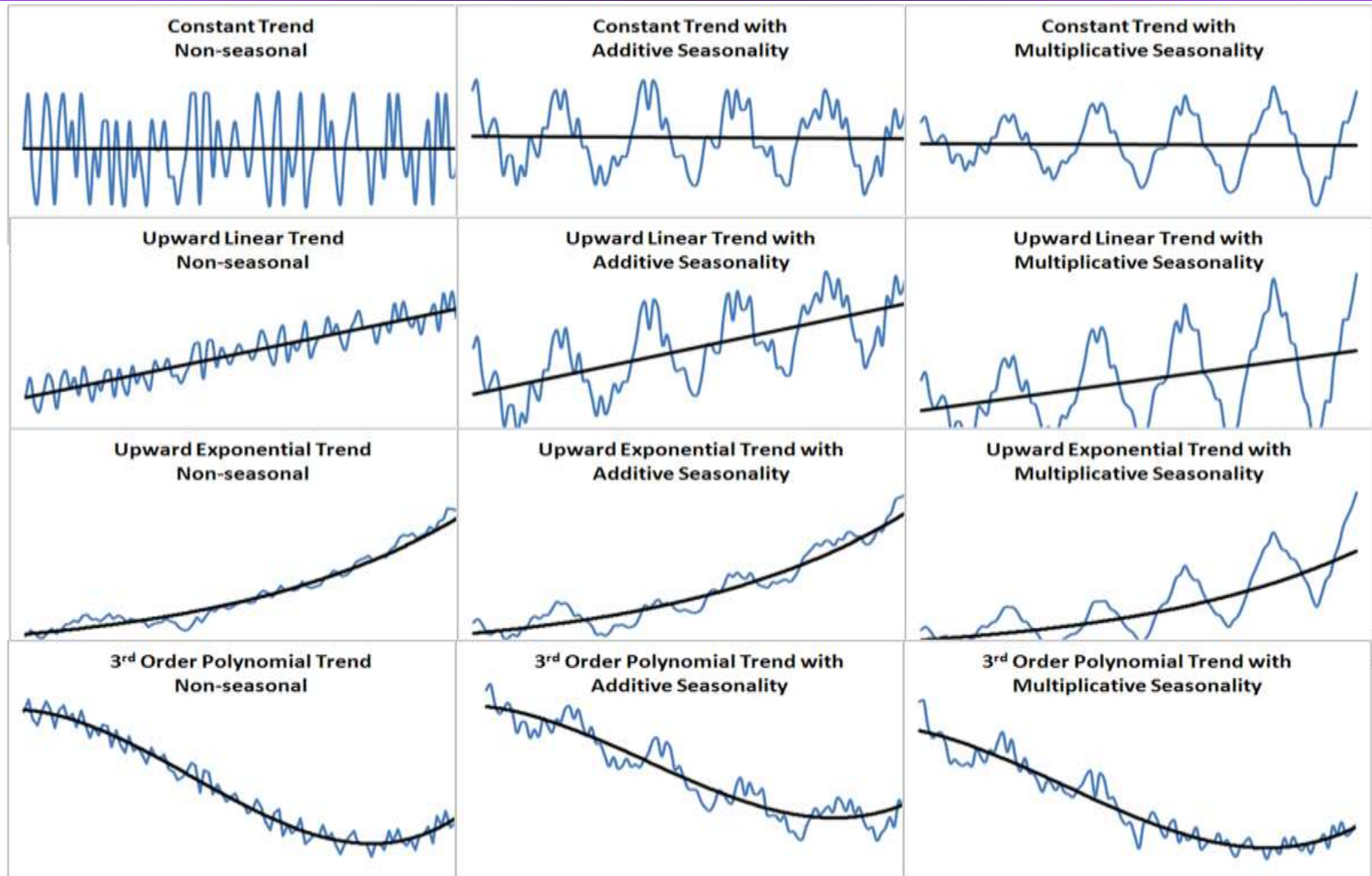
Time Plot for Amtrak Ridership



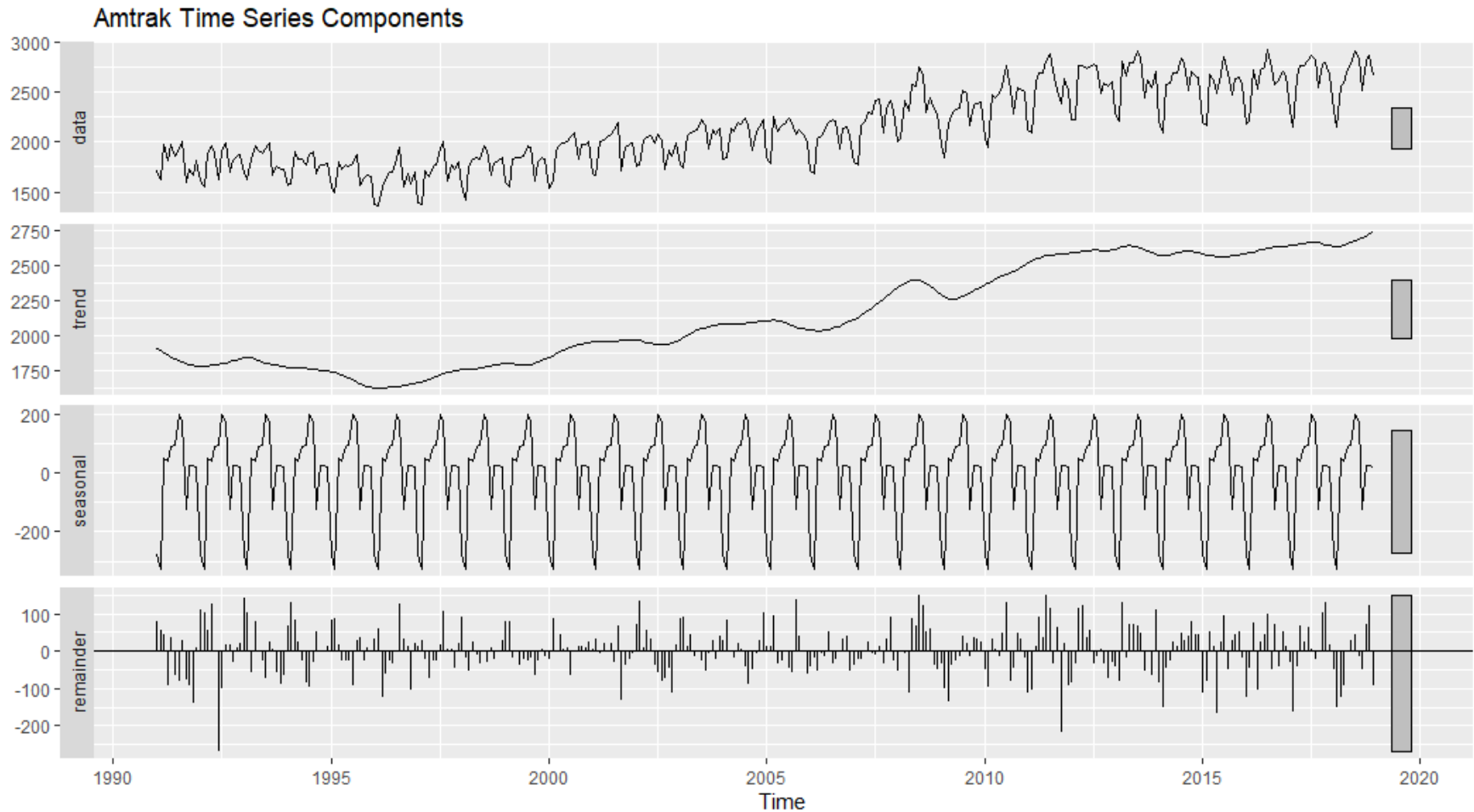
Time Series Components

- **Systematic part components**
 - **Level (stationary)** describes the average value of the series
 - **Trend** is the upward or downward change in the series from one period to the next
 - **Seasonality** is a short-term cyclical behavior observed several times in a time series
- **Non-systematic part component**
 - **Noise (randomness)** – random variations resulting from measurement error or other causes not accounted for
- Time series components are commonly considered either *additive* or *multiplicative*
 - **Additive components**
 - » $Y_t = \text{Level} + \text{Trend} + \text{Seasonality} + \text{Noise}$
 - **Multiplicative components**
 - » $Y_t = \text{Level} \times \text{Trend} \times \text{Seasonality} \times \text{Noise}$
- While *not all time series have trend and/or seasonality*, *all series have a level*
- Forecasting methods attempt to isolate the systematic part and quantify noise (randomness)
 - Systematic part is used to generate point forecasts
 - Noise (randomness) helps assess the uncertainty associated with point forecasts

Charts of Time Series Components



Amtrak Time Series Components Using stl() Function in R



Autocorrelation

- **Autocorrelation** represents the correlation between a random variable (time series data) itself and the same variable lagged one or more periods
- **Autocorrelation coefficient r_k** is a measure of autocorrelation between a variable itself and the same variable lagged k periods:

$$r_k = \frac{\sum_{t=k+1}^n (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^n (Y_t - \bar{Y})^2}$$

where

r_k = autocorrelation coefficient for a lag of k periods ($k = 1, 2, 3, \dots, 12, \dots$)

$\bar{Y}$ = mean of the values of the series

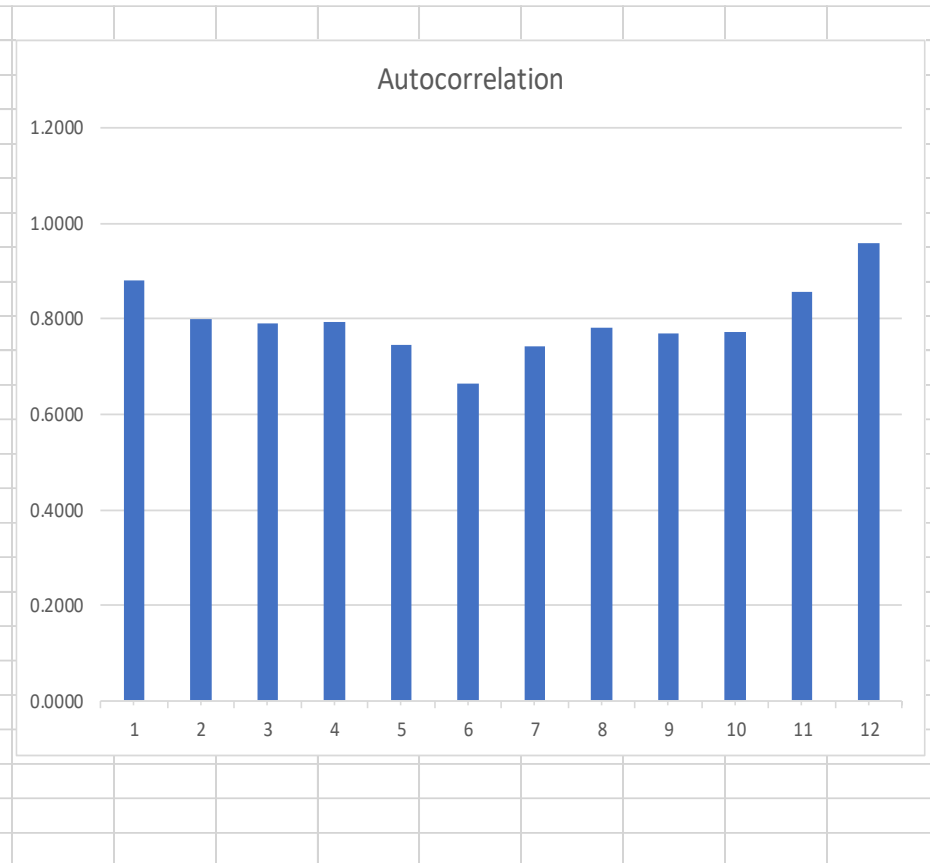
Y_t = observation in time period t

Y_{t-k} = observation k time periods earlier or at time period $t-k$

- If a time series is **random**, the autocorrelations for any lag are close to zero
 - The successive values of a time series are not related to each other

Autocorrelation Coefficients (Excel)

Period	Month	Ridership	Lag 1	Lag 2	Lag 12	Lag k	Autocorrelation
1	1/1/1991	1708.917				1	0.8813
2	2/1/1991	1620.586	1708.917			2	0.7993
3	3/1/1991	1972.715	1620.586	1708.917		3	0.7897
4	4/1/1991	1811.665	1972.715	1620.586		4	0.7949
5	5/1/1991	1974.964	1811.665	1972.715		5	0.7471
6	6/1/1991	1862.356	1974.964	1811.665		6	0.6645
7	7/1/1991	1939.860	1862.356	1974.964		7	0.7424
8	8/1/1991	2013.264	1939.860	1862.356		8	0.7817
9	9/1/1991	1595.657	2013.264	1939.860		9	0.7688
10	10/1/1991	1724.924	1595.657	2013.264		10	0.7717
11	11/1/1991	1675.667	1724.924	1595.657		11	0.8561
12	12/1/1991	1813.863	1675.667	1724.924		12	0.9592
13	1/1/1992	1614.827	1813.863	1675.667	1708.917		
14	2/1/1992	1557.088	1614.827	1813.863	1620.586		
15	3/1/1992	1891.223	1557.088	1614.827	1972.715		
16	4/1/1992	1955.981	1891.223	1557.088	1811.665		
17	5/1/1992	1884.714	1955.981	1891.223	1974.964		
18	6/1/1992	1623.042	1884.714	1955.981	1862.356		
19	7/1/1992	1903.309	1623.042	1884.714	1939.860		
20	8/1/1992	1996.712	1903.309	1623.042	2013.264		
21	9/1/1992	1703.897	1996.712	1903.309	1595.657		
22	10/1/1992	1810.000	1703.897	1996.712	1724.924		
23	11/1/1992	1861.601	1810.000	1703.897	1675.667		
24	12/1/1992	1875.122	1861.601	1810.000	1813.863		

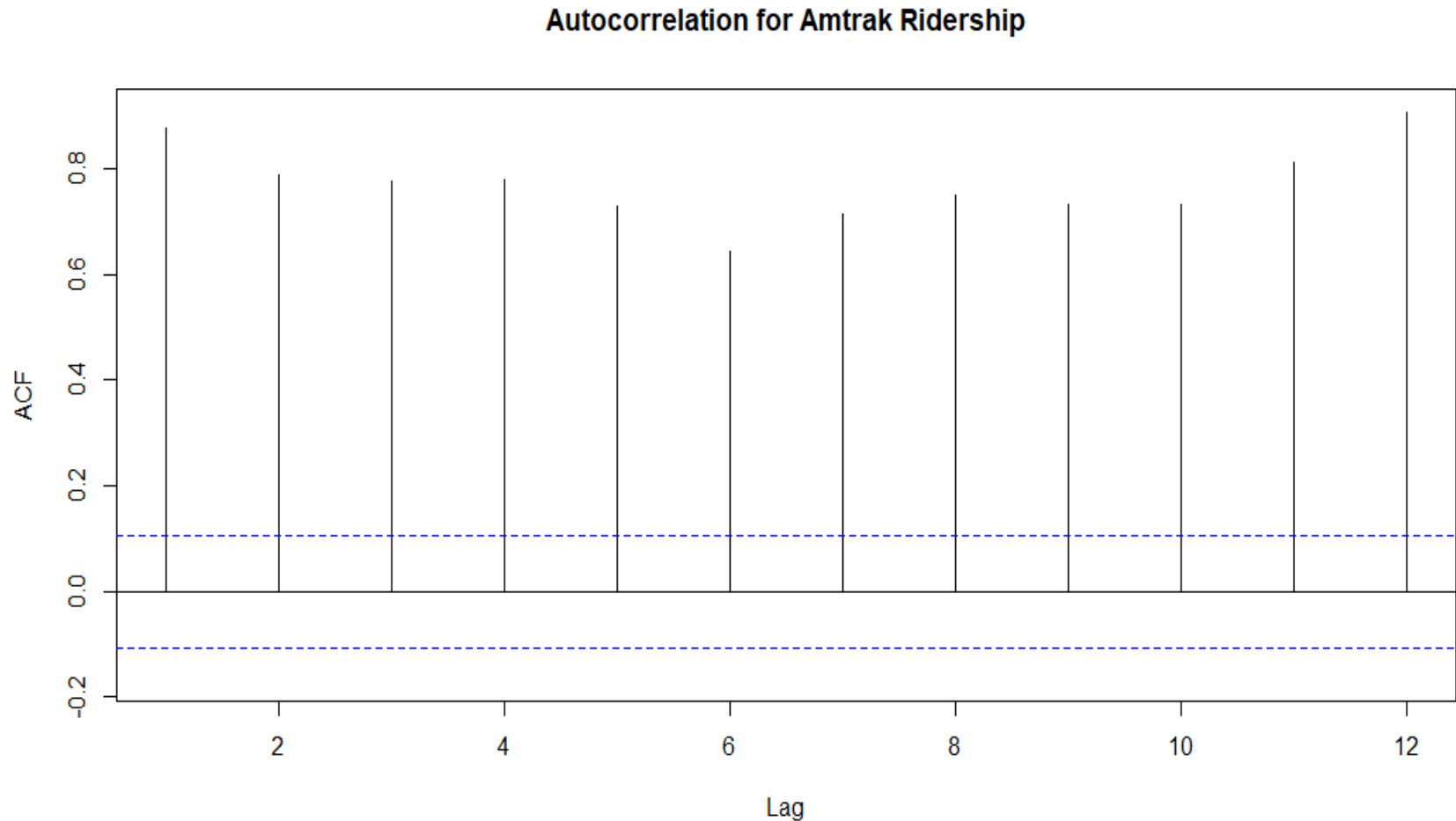


Autocorrelation Function (Acf)

- **Autocorrelation function (Acf)** identifies and plots multiple autocorrelation coefficients for various lags of a time series data
 - It also calculates and plots interval (horizontal lines) to identify significant autocorrelation
- **Correlogram** – is a plot of autocorrelation (autocorrelation coefficients) for various lags
- Correlogram can be used to analyze the autocorrelation coefficients for different time lags and answer the following questions:
 - Does the time series data have a trend component?
 - Does the time series data have a seasonal component?
 - Does the time series data have a level component?
 - Is the data random?



Correlogram Using Acf() Function in R



Analyzing Patterns with Autocorrelation

■ *Series with trends*

- Positive autocorrelation coefficients are significantly greater from zero for the first *lag 1* or first several periods (*lag 1, lag 2, etc.*) and then may gradually drop down, sometimes to zero or negative values

■ *Series with seasonality*

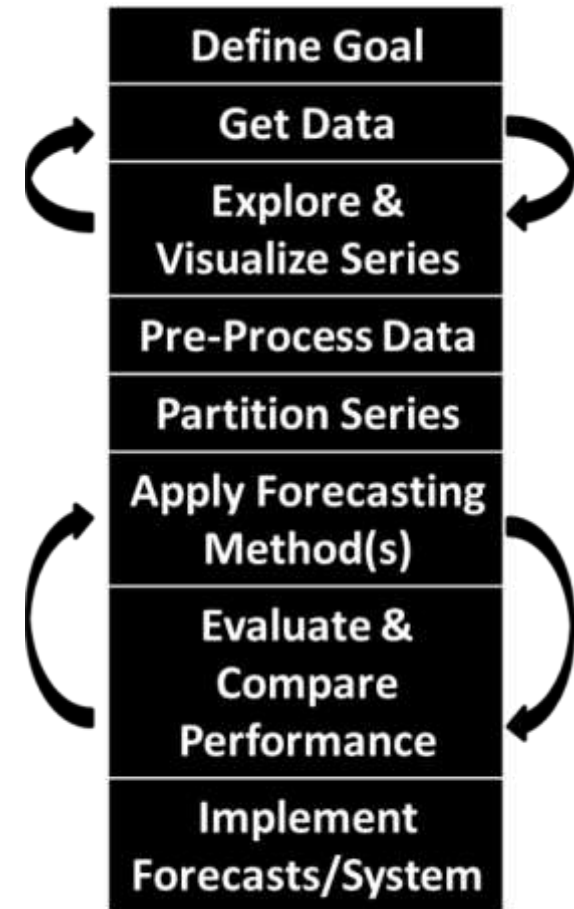
- Autocorrelation coefficients are significant at the appropriate time lag: *lag 4* for quarterly data, *lag 12* for monthly data

■ *Series with level (stationary) pattern*

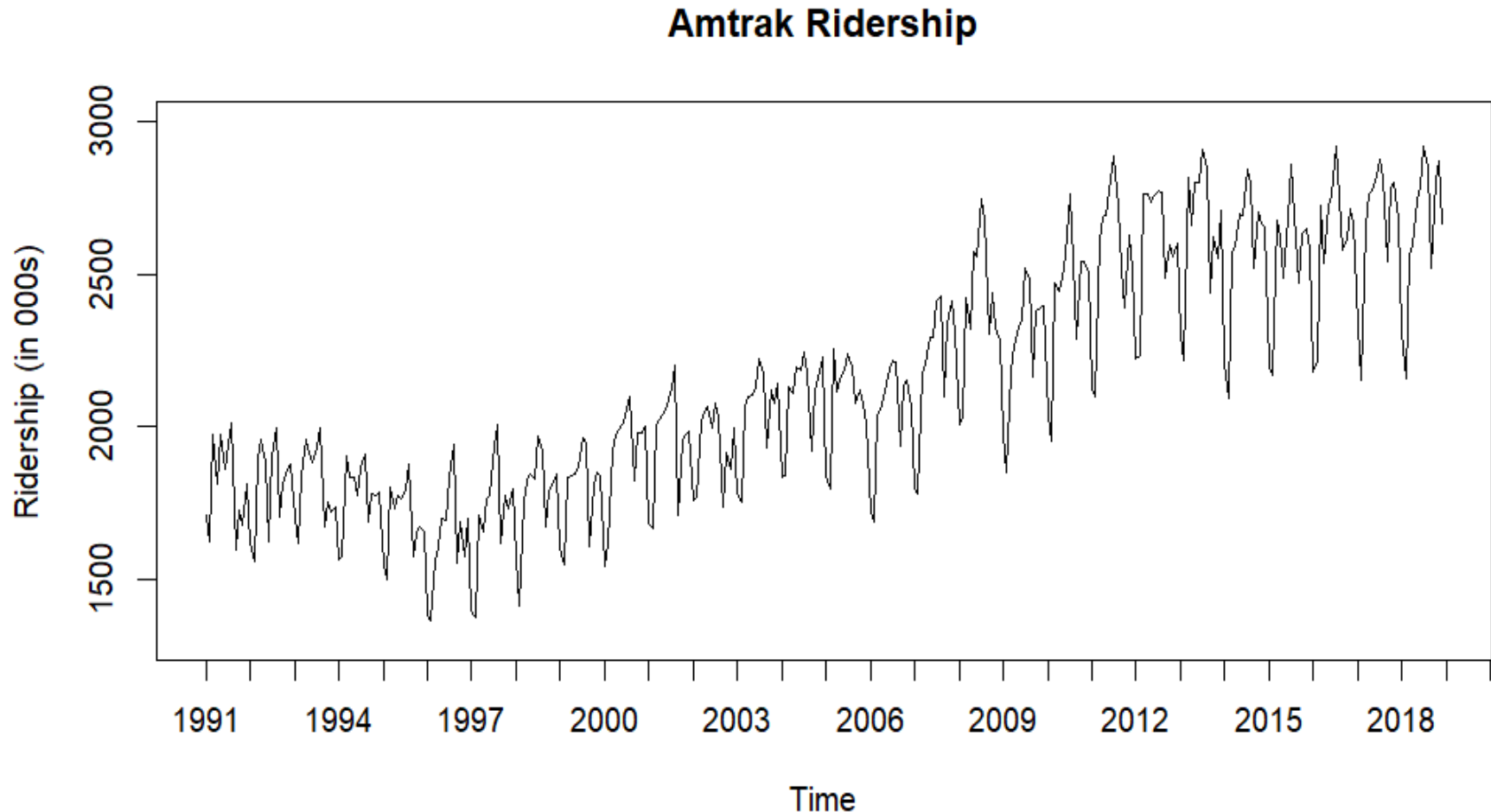
- Autocorrelation coefficients decline fairly rapidly, generally after the second- to fourth-time lag

Step 3: Explore & Visualize Series

- **Visualize series** step is used as an effective initial step for *characterizing the nature of time series*
- May detect *potential issues with the data*
 - *Extreme values*
 - *Unequal spacing*
 - *Missing values*
- **Time plot** is most basic but very informative plot for visualizing a time series
 - Line chart of the series values ($y_1, y_2, \dots$) over time ($t = 1, 2, \dots$) with temporal labels (e.g., calendar date) on the horizontal axis

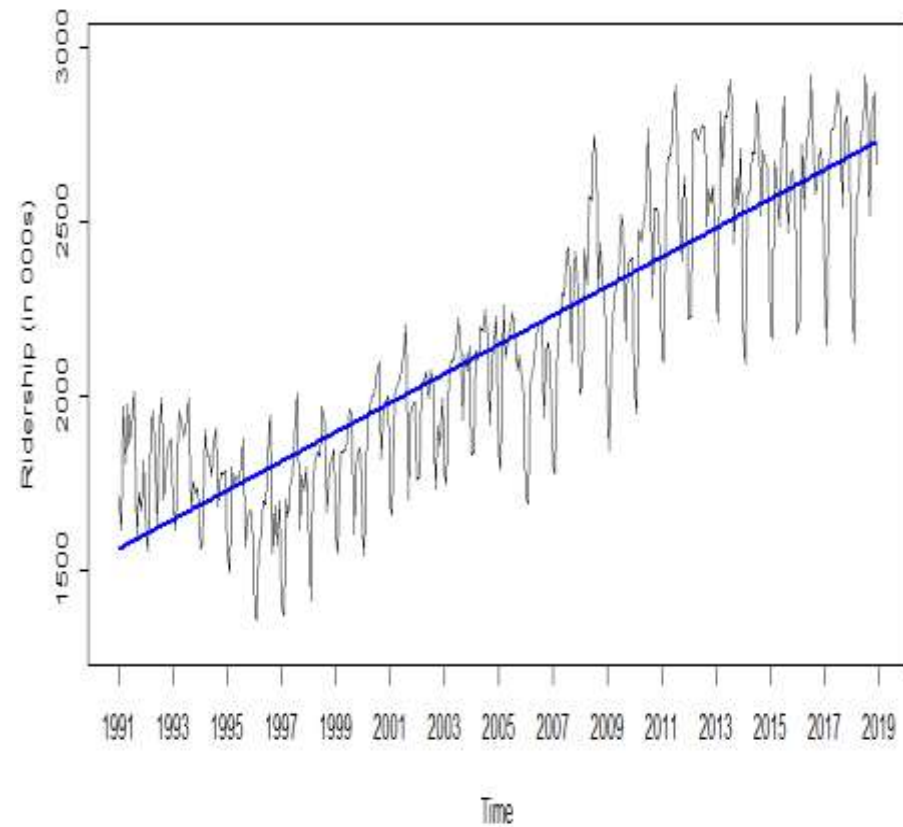


Time Plot for Amtrak Ridership Using plot() Function in R

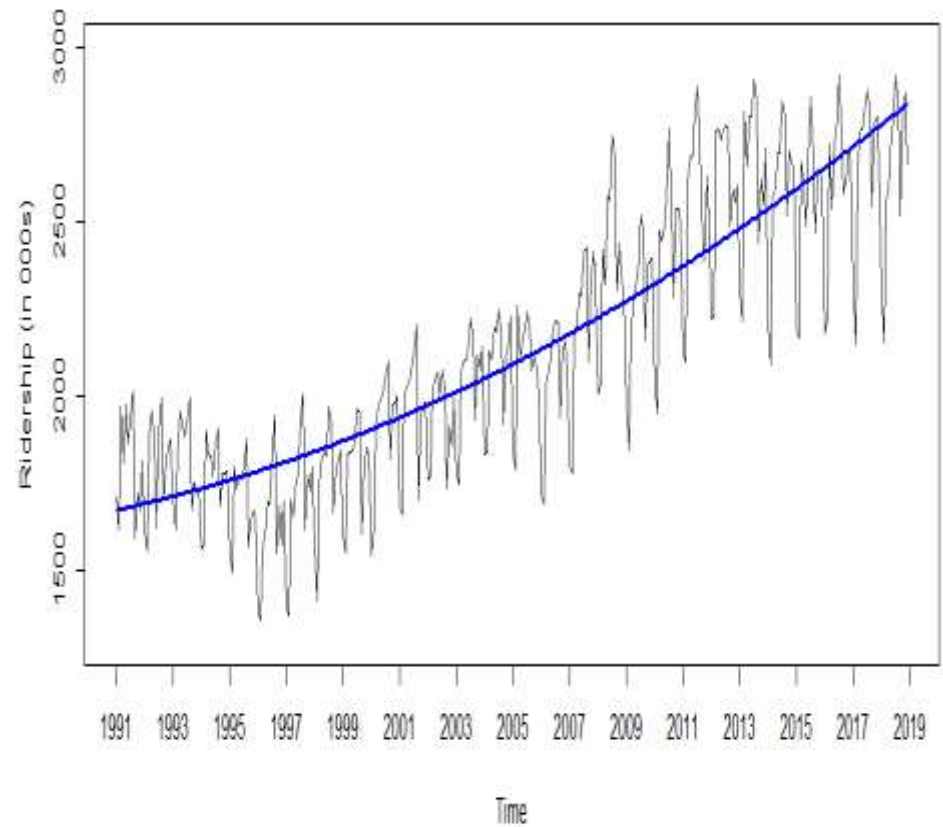


Time Plot with Trendline

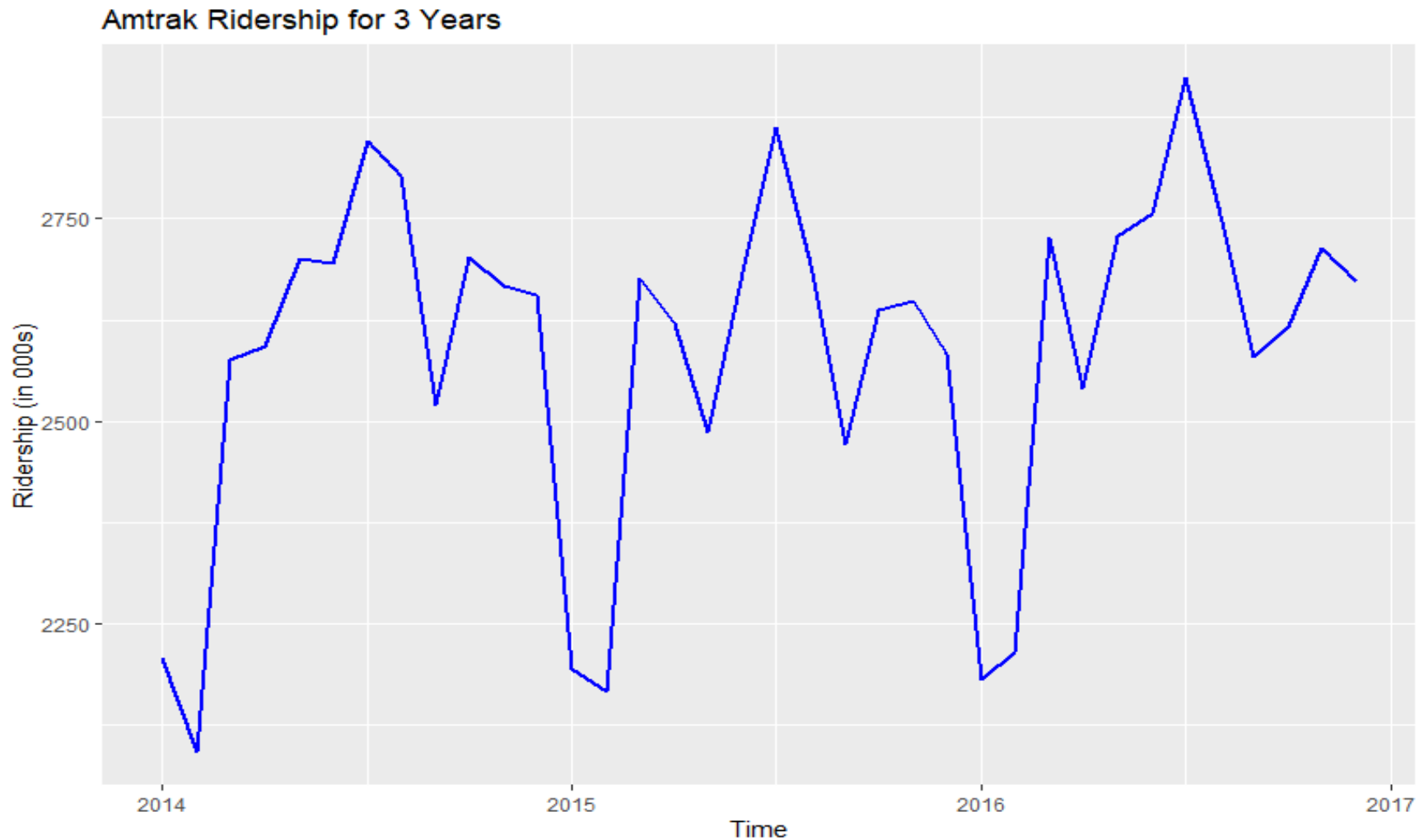
Amtrak Ridership with Linear Trend



Amtrak Ridership with Quadratic Trend



Zoom-In Time Plot Using autoplot() Function in R



Step 4: Pre-Process Data

- ***Pre-process data*** is used to detect potential issues and challenges with data
 - *Extreme values (outliers)*
 - *Data entry errors*
 - *Missing values*
 - *Unequal spacing*
 - *Irrelevant periods*
 - *Time span (how far back?)*
- May require *data cleaning* to remove time series issues

